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Enhancing Rabies Epidemic Modeling with Neural Networks and Fractional Calculus

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Abstract

Rabies remains a major public health concern, particularly in regions where dog-mediated transmission sustains human infection risk. Classical rabies models often rely on integer-order dynamics and overlook the long incubation periods and delayed behavioral responses that characterize the disease. Moreover, neural-network-based surrogate frameworks have rarely been developed for multi-host rabies systems incorporating biologically realistic incidence mechanisms and fractional memory effects. In this study, we propose an eight-compartment human–dog rabies transmission model governed by the Atangana–Baleanu–Caputo (ABC) fractional derivative, which captures nonlocal memory through a non-singular Mittag–Leffler kernel. Transmission between hosts is modeled using a harmonic-mean incidence rate to reflect saturation effects in dog–human and dog–dog contacts. To efficiently approximate the resulting fractional dynamics, we develop a Levenberg–Marquardt-based deep neural network (LMB–DNN) surrogate trained on reference solutions generated by a fractional Adams–Bashforth–Moulton (ABM) predictor–corrector scheme. The neural surrogate accurately reproduces the fractional model dynamics across all eight state variables, achieving mean squared errors in the range 10^{-8} – 10^{-6} , absolute errors below 10^{-5} , and regression coefficients close to unity. A systematic investigation of the fractional order $\alpha \in (0.85, 0.95]$ reveals that memory effects significantly influence outbreak timing, peak magnitude, and persistence, with moderate memory ($\alpha \approx 0.90$ – 0.95) producing smoother trajectories consistent with rabies incubation delays. Results from the sensitivity analysis indicate that parameters associated with dog-to-dog transmission and the incubation process exert the strongest influence on the basic reproduction number, suggesting that effective rabies control should primarily focus on dogs. The proposed ABC fractional LMB–DNN framework provides a computa-

tionally efficient and biologically meaningful tool for analyzing rabies dynamics with memory effects. The approach offers a practical alternative to repeated fractional simulations, supports rapid scenario exploration, and establishes a foundation for future integration of epidemiological data, uncertainty quantification, and adaptive control strategies in rabies modeling.

Keywords: Rabies transmission modeling; infectious-disease spread; numerical simulation; neural-network methods; Levenberg–Marquardt training; fractional calculus.

1. Introduction

Rabies is an acute viral infection of the central nervous system that affects mammals, humans included. Transmission most often follows the bite of an infectious animal, commonly a dog or bat, although exposure can also occur when saliva from an infected animal contacts broken skin or mucous membranes (eyes, nose, or mouth) [1, 2]. Early clinical signs are typically nonspecific fever, headache, myalgias, and paresthesia or pruritus at the wound site and may progress to seizures, hallucinations, and paralysis. After symptom onset, the disease is almost uniformly fatal. Nevertheless, timely wound care and post-exposure prophylaxis (PEP) with vaccine, with or without rabies immunoglobulin, can prevent illness.

Rabies occurs worldwide, its impact is greatest in Africa and Asia, where dog-mediated transmission accounts for the overwhelming majority of human deaths. Global estimates suggest tens of thousands of fatalities each year, with children and underserved populations at disproportionate risk. In high income regions, wildlife reservoirs such as foxes, raccoons, and bats sustain transmission and occasionally spill over to domestic animals and humans. Historical reports document human animal conflicts related to rabies stretching back millennia [3, 4].

Mathematical modelling has substantially advanced our understanding of rabies dynamics and the expected effects of control strategies [5, 6, 7]. Numerous studies have quantified vaccination impacts in wildlife reservoirs, including raccoons [8, 9], often formulating vaccination as a control input [10, 11]; related approaches have also been explored in China with setting-specific modifications [12]. Beyond classical systems, fractional-calculus methods have been successfully applied to epidemic models [13, 14]. Fractional operators can encode memory and nonlocal effects, often improving fit to surveillance data and capturing biological persistence. The Caputo operator is widely used because of its favorable analytical and numerical properties and its ability to model nonsingular, nonlocal dynamics [15]. More generally, fractional derivatives can be derived from the Riemann–Liouville fractional integral, extending standard differentiation and

integration to non-integer orders [18, 19]. These tools have seen broad application in mathematical biology, including diffusion in porous media, hemodynamics, and pathogen spread. The literature includes both singular and nonsingular kernels [21, 22] and comprehensive analyses of their implementation [23, 24].

Among nonsingular-kernel operators, the Atangana–Baleanu (AB) derivative introduced in 2016 [15] generalizes Caputo-type formulations and has shown advantages for representing diffusion-like and epidemiological processes. Applications span infectious-disease transmission, tumor-growth models, and pollutant transport in porous media, where nonlocality and memory effects are important. We formulate the rabies dynamics using the ABC fractional derivative to incorporate incubation-driven and behavior-driven memory effects in a physically meaningful way. While the classical Caputo operator is widely used in epidemiology, its power-law kernel is singular near $t = 0$, which can overweight very recent history and may lead to stiffness in simulations. In contrast, the ABC operator replaces the singular kernel by a non-singular, nonlocal Mittag–Leffler kernel, yielding a flexible fading-memory mechanism that has been reported as advantageous for diffusion-like and epidemiological processes where hereditary effects are important [16, 17]. This makes ABC a suitable choice for rabies, where long and variable incubation periods and delayed responses imply that past exposure continues to influence present infection progression.

Despite substantial advances in rabies and epidemic modeling, existing studies offer limited understanding of memory-dependent multi-host transmission mechanisms, particularly for dog to human interactions modeled through biologically meaningful incidence structures. Most classical epidemic models, as well as many fractional formulations, either consider single-host populations or employ integer-order dynamics, which are unable to adequately represent the long incubation periods, delayed behavioral responses, and persistent effects of exposure observed in viral infections. Recent works have demonstrated that fractional-order operators, combined with deep neural network–based numerical solvers, provide an effective framework for incorporating nonlocal memory effects and reducing the computational burden associated with fractional dynamics [25, 26, 27]. In particular, learning-assisted fractional models have been successfully applied to poliomyelitis, rabies, and hepatitis B dynamics, highlighting their capability to capture complex temporal behaviors while maintaining numerical efficiency [25, 26, 27]. The extension of such fractional learning-based approaches to two-host rabies systems, especially those incorporating realistic incidence formulations, remains largely unexplored. These gaps motivate the development of a modeling framework that integrates fractional calculus based memory effects with computationally efficient neural solvers for multi-host rabies transmission dynamics.

In this work, we address this gap by formulating an eight-compartment human–dog rabies model governed by the ABC derivative and by employing a LM trained neural network to learn its nonlinear time evolution. This framework provides a mathematically novel treatment of harmonic-mean transmission under fractional memory, a computationally efficient alternative to repeated fractional simulations, and an epidemiologically relevant platform for analyzing intervention strategies in settings where incubation delays and historical exposure significantly influence disease progression. We adopt a modified version of the rabies model of Eti Dwi et al. [28], following related developments in [29, 30]. The transmission term is specified using a harmonic-mean incidence, which is well suited for rapid spread between infectious dogs and susceptible humans when the interacting populations are measured in comparable units. The human population is partitioned into susceptible (S_P), exposed (E_P), infectious (I_P), and recovered (R_P) classes; the dog population is similarly divided into S_D , E_D , I_D , and R_D . The total population sizes are

$$N_P = S_P + E_P + I_P + R_P, \quad N_D = S_D + E_D + I_D + R_D.$$

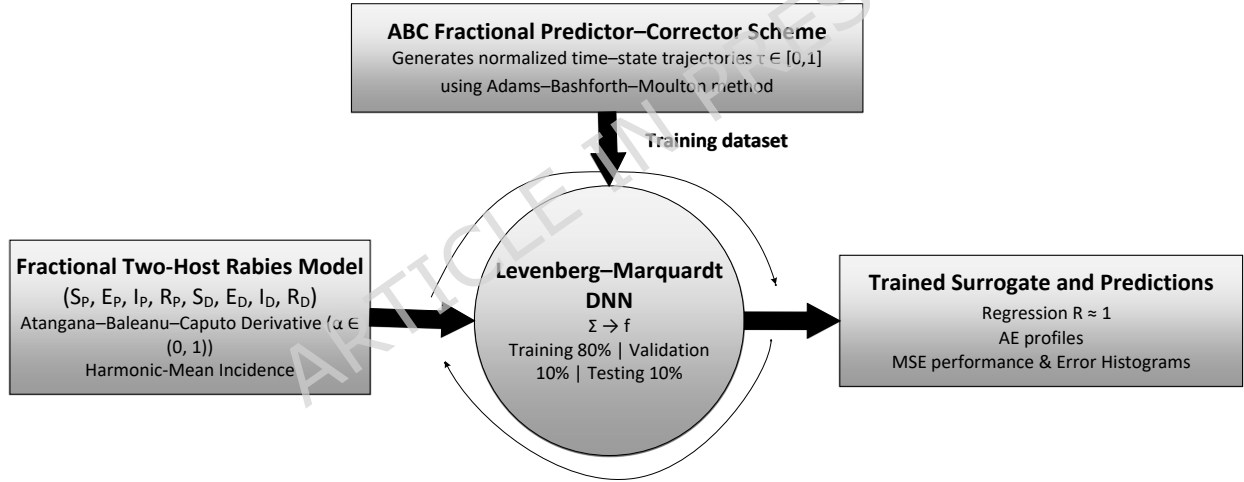


Figure 1: Comprehensive overview illustrating the performance and accuracy of the proposed fractional-order rabies model and its neural learning architecture across all tested scenarios.

Mathematical models play a fundamental, powerful, and significant role in understanding infectious disease dynamics and guiding infection control practice in epidemiology. We study a two-host rabies transmission model that partitions the human population into susceptible S_P , exposed E_P , infected I_P , and recovered R_P , and the dog population into S_D , E_D , I_D , and R_D . Transmission is modeled via a harmonic mean incidence, e.g., $\frac{2S_P I_D}{S_P + I_D}$ for dog $\rightarrow$ human infections, $\frac{2S_D I_P}{S_D + I_P}$ for dog $\rightarrow$ dog spread, together with re-

cruitment, vaccination, waning immunity, removal, and natural mortality in both hosts. To capture memory and nonlocal effects, the time evolution employs the ABC fractional derivative of order $\alpha \in (0, 1)$, yielding an eight-dimensional, nonlinear fractional system.

The present work aims to solve and analyze this fractional rabies model with a LMB deep neural network. A reference dataset is generated for several parameter cases using a fractional ABM scheme; a one-hidden-layer feed-forward network is then trained on the normalized time domain. Model fidelity is assessed via training/validation/testing splits, absolute-error (AE) curves on a semilog scale, regression fits, error histograms, and MSE trajectories, demonstrating the network's ability to approximate the ABC fractional rabies dynamics across cases, as shown in Figure 1. This framework extends DNN-based epidemic modeling by addressing an eight-compartment human-dog rabies system with fractional dynamics and harmonic-mean incidence:

$$\left\{ \begin{array}{ll} {}^{ABC}D_t^\alpha S_P = \Lambda_P - B_{DP} \left(\frac{2S_P I_D}{S_P + I_D} \right) - m_P S_P + w_P R_P - \mu_P S_P, & S_P(0) = i_1, \\ {}^{ABC}D_t^\alpha E_P = B_{DP} \left(\frac{2S_P I_D}{S_P + I_D} \right) - (\varepsilon_P + \mu_P) E_P, & E_P(0) = i_2, \\ {}^{ABC}D_t^\alpha I_P = \varepsilon_P E_P - (\alpha_P + \mu_P) I_P, & I_P(0) = i_3, \\ {}^{ABC}D_t^\alpha R_P = m_P S_P - (w_P + \mu_P) R_P, & R_P(0) = i_4, \\ {}^{ABC}D_t^\alpha S_D = \Lambda_D - B_{DD} \left(\frac{2S_D I_D}{S_D + I_D} \right) - m_D S_D + w_D R_D - \mu_D S_D, & S_D(0) = i_5, \\ {}^{ABC}D_t^\alpha E_D = B_{DD} \left(\frac{2S_D I_D}{S_D + I_D} \right) - (\varepsilon_D + \mu_D) E_D, & E_D(0) = i_6, \\ {}^{ABC}D_t^\alpha I_D = \varepsilon_D E_D - (\alpha_D + \mu_D) I_D, & I_D(0) = i_7, \\ {}^{ABC}D_t^\alpha R_D = m_D S_D - (w_D + \mu_D) R_D, & R_D(0) = i_8. \end{array} \right. \quad (1.1)$$

2. Model statement and context

The mathematical system in Equation (1.1) describes an eight-compartment, two-host rabies transmission model that accounts for both human and dog populations. The dynamics are governed by the ABC fractional derivative of order $\alpha \in (0, 1)$, which captures memory and hereditary effects in infection processes.

The state vector $(S_P, E_P, I_P, R_P, S_D, E_D, I_D, R_D)$ denotes the susceptible (S), exposed (E), infected (I), and recovered/vaccinated (R) classes in humans and dogs, respectively. Transmission is modeled via *harmonic-mean incidence*: $\frac{2S_P I_D}{S_P + I_D}$ for dog $\rightarrow$ human infection, $\frac{2S_D I_D}{S_D + I_D}$ for dog $\rightarrow$ dog spread, while de-

mographic and control processes (recruitment, vaccination, waning of immunity, and natural mortality) act within each host population. The model parameters are Λ_P, Λ_D (recruitment/birth rates), μ_P, μ_D (natural death rates), α_P, α_D (removal/isolation rates), m_P, m_D (vaccination rates), B_{DP}, B_{DD} (transmission coefficients), $\varepsilon_P, \varepsilon_D$ (incubation rates), w_P, w_D (waning rates of immunity). The initial conditions are

$$S_P(0) = i_1, E_P(0) = i_2, I_P(0) = i_3, R_P(0) = i_4, S_D(0) = i_5, E_D(0) = i_6, I_D(0) = i_7, R_D(0) = i_8.$$

Classical SIR-type epidemic models have been widely used to study infection spread, control strategies, and recovery dynamics [32]. These frameworks built on mechanistic assumptions have been extended to incorporate optimal control [33], fractional-order SEIR systems [34], and vaccination-focused strategies [35, 36, 37]. Despite substantial progress in both analytical and numerical modeling, to our knowledge a LMB-DNN has not previously been applied to an ABC-fractional, two-host rabies system with harmonic-mean incidence. This work addresses that gap. Several ANN-based approaches have been proposed for solving epidemic models, including rabies-type systems. However, most existing studies focus on integer-order dynamics, single-host formulations, or standard bilinear incidence rates, and primarily aim at improving numerical accuracy of the solution. In contrast, the present work integrates ABC fractional dynamics, a harmonic-mean incidence rate, and a two-host (human–dog) rabies structure within a unified framework. Moreover, unlike commonly used gradient-based optimizers (e.g., Adam), we employ a LMB–DNN to construct an efficient surrogate for the fractional-order model. This combination allows us to extract both epidemiological insight and computational insight, rather than merely improving numerical accuracy. In the present study, the neural-network surrogate is trained and validated against reference trajectories generated by a high-accuracy fractional ABM predictor–corrector scheme. This choice is intentional and reflects the methodological focus of the work: assessing the ability of the proposed LMB–DNN framework to accurately and efficiently learn the dynamics of an ABC fractional rabies model. We propose and analyze a neural model framework for the fractional rabies model with the following key advances:

- We design a LMB–DNN that captures the nonlinear, eight-dimensional ABC-fractional dynamics of the coupled human–dog system.
- The network is trained on datasets produced by a fractional time integrator across multiple parameter regimes and fractional orders α , spanning diverse epidemiological and control scenarios.
- Accuracy is quantified against a fractional reference solver using regression analysis, MSE trajectories, semilog AE profiles, and residual histograms for all eight states.

- The LMB–DNN exhibits numerical stability, straightforward implementation, and adaptability to other multi-host and fractional epidemic systems with control and memory effects.

3. Mathematical Analysis of the Model

In this section, we investigate the mathematical properties of the proposed ABC fractional-order rabies model given in system (1.1). In particular, we establish the positivity and boundedness of solutions, derive the disease-free equilibrium, compute the basic reproduction number, and analyze the local stability of the disease-free equilibrium (DFE).

3.1. Positivity of Solutions

Theorem 3.1. *Let the initial conditions of system (1.1) satisfy*

$$S_P(0), E_P(0), I_P(0), R_P(0), S_D(0), E_D(0), I_D(0), R_D(0) \geq 0.$$

Then, the solutions of system (1.1) remain non-negative for all $t > 0$.

Proof. We show that the vector field of system (1.1) points inward on the boundary of the non-negative orthant. Consider the susceptible human population. If $S_P(t) = 0$, then from the first equation of system (1.1) we obtain ${}^{ABC}D_t^\alpha S_P = \Lambda_P + w_P R_P \geq 0$, which implies that $S_P(t)$ cannot cross into the negative region. Similarly, if $E_P(t) = 0$, then ${}^{ABC}D_t^\alpha E_P = \frac{2B_{DP}S_P I_D}{S_P + I_D} \geq 0$, and if $I_P(t) = 0$, we have ${}^{ABC}D_t^\alpha I_P = \varepsilon_P E_P \geq 0$. Analogous arguments hold for the remaining state variables R_P, S_D, E_D, I_D , and R_D . In each case, whenever a compartment variable vanishes, its corresponding ABC fractional derivative is non-negative. Therefore, no solution trajectory starting in $\mathbb{R}_+^8$ can leave the non-negative region. Hence, all solutions of system (1.1) remain non-negative for all $t > 0$. $\square$

3.2. Boundedness and Invariant Region

Theorem 3.2. *All solutions of system (1.1) are uniformly bounded. In particular, for all $t > 0$, the total human and dog populations satisfy $0 \leq N_P(t) \leq \frac{\Lambda_P}{\mu_P}$, $0 \leq N_D(t) \leq \frac{\Lambda_D}{\mu_D}$.*

Proof. Let $N_P(t) = S_P(t) + E_P(t) + I_P(t) + R_P(t)$, $N_D(t) = S_D(t) + E_D(t) + I_D(t) + R_D(t)$ denote the total human and dog populations, respectively. Summing the first four equations of system (1.1) yields ${}^{ABC}D_t^\alpha N_P(t) = \Lambda_P - \mu_P N_P(t)$, which is a linear fractional differential equation. Its solution satisfies $N_P(t) \leq \max\left\{N_P(0), \frac{\Lambda_P}{\mu_P}\right\}$, $\forall t > 0$. Similarly, summing the last four equations of system (1.1) gives ${}^{ABC}D_t^\alpha N_D(t) =$

$\Lambda_D - \mu_D N_D(t)$, from which it follows that $N_D(t) \leq \max \left\{ N_D(0), \frac{\Lambda_D}{\mu_D} \right\}$, $\forall t > 0$. Therefore, all solution trajectories of system (1.1) eventually enter and remain in the positively invariant region

$$\Omega = \left\{ (S_P, \dots, R_D) \in \mathbb{R}_+^8 : N_P \leq \frac{\Lambda_P}{\mu_P}, N_D \leq \frac{\Lambda_D}{\mu_D} \right\}.$$

Hence, system (1.1) is uniformly bounded in Ω . $\square$

3.3. Disease-Free Equilibrium and Basic reproduction number

In mathematical models of infectious diseases, R_0 is a commonly used term to describe the basic reproduction number. This number represents the average number of secondary infections that one infected individual would produce in a completely susceptible population. If $R_0 < 1$, then the disease is not able to sustain itself in the population and will eventually die out. On the other hand, if $R_0 > 1$, the disease will be able to spread and become endemic. For the value of R_0 in model (1.1), the disease-free equilibrium, denoted by E^0 , is $E^0 = (S_{P0}, E_{P0}, I_{P0}, R_{P0}, S_{D0}, E_{D0}, I_{D0}, R_{D0})$,

$$= \left(\frac{(w_P + \mu_P)\Lambda_P}{\mu_P(w_P + \mu_P + m_P)}, 0, 0, \frac{m_P\Lambda_P}{\mu_P(w_P + \mu_P + m_P)}, \frac{(w_D + \mu_D)\Lambda_D}{\mu_D(w_D + \mu_D + m_D)}, 0, 0, \frac{m_D\Lambda_D}{\mu_D(w_D + \mu_D + m_D)} \right). \quad (3.1)$$

Let (E_P, I_P, E_D, I_D) be the infected compartments. Then system (1.1) reduces to

$$\begin{cases} \frac{dE_P}{dt} = B_{DP} \left(\frac{2S_P I_D}{S_P + I_D} \right) - (\varepsilon_P + \mu_P)E_P, \\ \frac{dI_P}{dt} = \varepsilon_P E_P - (\alpha_P + \mu_P)I_P, \\ \frac{dE_D}{dt} = B_{DD} \left(\frac{2S_D I_D}{S_D + I_D} \right) - (\varepsilon_D + \mu_D)E_D, \\ \frac{dI_D}{dt} = \varepsilon_D E_D - (\alpha_D + \mu_D)I_D. \end{cases} \quad (3.2)$$

For system (3.2), the Jacobian matrix is

$$J^* = \begin{pmatrix} -(\varepsilon_P + \mu_P) & 0 & 0 & 2B_{DP} \\ \varepsilon_P & -(\alpha_P + \mu_P) & 0 & 0 \\ 0 & 0 & -(\varepsilon_D + \mu_D) & 2B_{DD} \\ 0 & 0 & \varepsilon_D & -(\alpha_D + \mu_D) \end{pmatrix}. \quad (3.3)$$

Arranging J^* such that $J^* = F - V$, we obtain

$$F = \begin{pmatrix} 0 & 0 & 0 & 2B_{DP} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2B_{DD} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (3.4)$$

and

$$-V = \begin{pmatrix} \varepsilon_P + \mu_P & 0 & 0 & 0 \\ -\varepsilon_P & \alpha_P + \mu_P & 0 & 0 \\ 0 & 0 & \varepsilon_D + \mu_D & 0 \\ 0 & 0 & -\varepsilon_D & \alpha_D + \mu_D \end{pmatrix}. \quad (3.5)$$

The dominant eigenvalue gives the basic reproduction number $R_0 = \frac{2B_{DD}\varepsilon_D}{(\varepsilon_D + \mu_D)(\alpha_D + \mu_D)}$.

3.4. Local stability

Theorem 3.3. *The disease-free equilibrium E^0 of system (1.1) is locally asymptotically stable if $R_0 < 1$.*

Proof. The Jacobian matrix of system (1.1) at E^0 is

$$J^{[0]} = \begin{pmatrix} -a_{11} & 0 & 0 & w_P & 0 & 0 & -2B_{DP} & 0 \\ 0 & -(\varepsilon_P + \mu_P) & 0 & 0 & 0 & 0 & 2B_{DP} & 0 \\ 0 & \varepsilon_P & -(\alpha_P + \mu_P) & 0 & 0 & 0 & 0 & 0 \\ m_P & 0 & 0 & -(w_P + \mu_P) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -(m_D + \mu_D) & 0 & -2B_{DD} & w_D \\ 0 & 0 & 0 & 0 & 0 & -a_{66} & 2B_{DD} & 0 \\ 0 & 0 & 0 & 0 & 0 & \varepsilon_D & -a_{77} & 0 \\ 0 & 0 & 0 & 0 & m_D & 0 & 0 & -a_{88} \end{pmatrix}. \quad (3.6)$$

Clearly, two eigenvalues of $J^{[0]}$ are $\lambda_1 = -(\alpha_P + \mu_P) < 0$, $\lambda_2 = -(\varepsilon_P + \mu_P) < 0$. The remaining eigenvalues are obtained from the characteristic equation

$$\begin{aligned} & (\lambda + (m_P + \mu_P))(\lambda + (w_P + \mu_P)(m_P + \mu_P) - m_P w_P) \times (\lambda + (m_D + \mu_D))(\lambda + (\varepsilon_D + \mu_D)) \\ & \times (\lambda + (\varepsilon_D + \mu_D)(\varepsilon_D + \mu_D) - 2\varepsilon_D B_{DD}) \times (\lambda + (m_D + \mu_D)(\varepsilon_D + \mu_D)(w_D + \mu_D) - m_D w_D(\varepsilon_D + \mu_D)) = 0. \end{aligned} \quad (3.7)$$

Thus, the eigenvalues are

$$\begin{aligned} \lambda_3 &= -(m_P + \mu_P) < 0, \\ \lambda_4 &= -(w_P + \mu_P)(m_P + \mu_P) + m_P w_P < 0, \\ \lambda_5 &= -(m_D + \mu_D) < 0, \\ \lambda_6 &= -(\varepsilon_D + \mu_D) < 0, \\ \lambda_7 &= (\varepsilon_D + \mu_D)(\varepsilon_D + \mu_D)(R_0 - 1) < 0, \quad \text{if } R_0 < 1, \\ \lambda_8 &= -(m_D + \mu_D)(\varepsilon_D + \mu_D)(w_D + \mu_D) + m_D w_D(\varepsilon_D + \mu_D) < 0. \end{aligned} \quad (3.8)$$

Hence, for $R_0 < 1$, all eigenvalues of the Jacobian matrix have negative real parts, and therefore the disease-free equilibrium E^0 is locally asymptotically stable. $\square$

Methodology

The proposed LMB–DNN framework is designed to approximate the dynamics of the eight-compartment, two-host rabies system representing human (P) and dog (D) populations governed by the ABC fractional derivative. The procedure consists of two primary stages: fractional data generation and neural training.

1. To build the supervised dataset, the ABC fractional model is integrated numerically across a normalized time interval $\tau \in [0, 1]$. A predictor–corrector procedure is used, combining the Grünwald–Letnikov formulation with fractional ABM steps. The input samples are simply the discrete time points $\{\tau_n\}$, and the corresponding targets are the eight state variables, $\mathbf{y}(\tau) = (S_P, E_P, I_P, R_P, S_D, E_D, I_D, R_D)^T$. Before training, all signals are rescaled to $[0, 1]$ to improve numerical conditioning and avoid instability during optimization.
2. The neural-network surrogate is built in MATLAB, using the `fitnet` function together with the LM routine (`trainlm`) for training. Nonlinear activation functions are used in the hidden layers, while the output layer remains linear so that the network can represent continuous state values. The dataset is split into training, validation, and testing sets in an 80–10–10 proportion using `divideParam`. Mean-squared error serves as the loss function, and an early-stopping check on validation error is used to guard against overfitting. Unless noted otherwise, a relatively small two-hidden-layer architecture is chosen to keep computations lightweight. All runs are executed through scripts rather than the `nftool` interface, allowing the experiments to be reproduced and enabling systematic sweeps over different parameter choices.

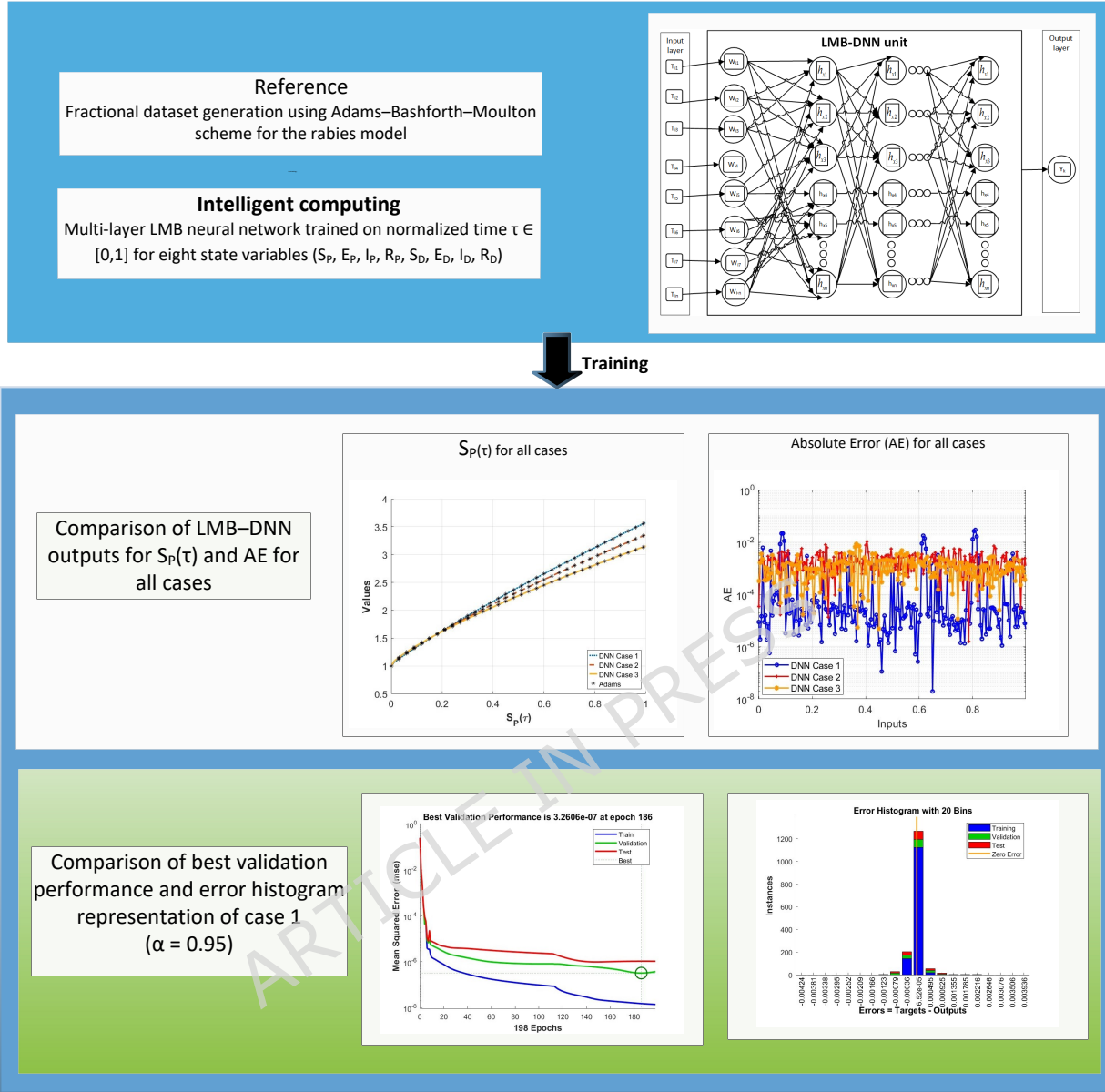


Figure 2: Workflow diagram of the LMB proposed DNN for the Rabies Model.

Figure 2 illustrates the complete computational workflow. A schematic of a four-neuron computation within the multilayer network is depicted in Figure 3. The case specific parameters including Λ_D and the fractional order α for each scenario are listed in Table 1. Training, validation, and testing metrics (best validation performance, gradient, damping parameter μ , epoch count, and wall-clock time) are summarized in Table 2.

The choice of a relatively shallow neural architecture is motivated by the regularity properties of the un-

derlying ABC fractional dynamics. The ABC derivative with a non-singular Mittag–Leffler kernel produces state trajectories that are continuous, smooth, and free of sharp discontinuities. As a result, the input–output mapping from normalized time to state variables is well behaved and does not require highly expressive deep architectures. In this setting, a compact feed-forward network with one or two hidden layers is sufficient to approximate the solution manifold accurately, consistent with universal approximation results for smooth functions. The number of hidden neurons was selected to balance approximation capability and computational efficiency. Sensitivity analysis shows that validation errors stabilize for moderate widths (approximately 20–30 neurons), with negligible improvement from further increases. This behavior arises from the smooth evolution of the fractional solutions, which limits the need for highly expressive neural networks. Adding excessive model capacity does not yield noticeable accuracy gains and can instead introduce over-parameterization. The use of nonlinear activations in the hidden layers is adequate for capturing the system’s nonlinear features, whereas a linear output layer preserves the continuity of the state variables.

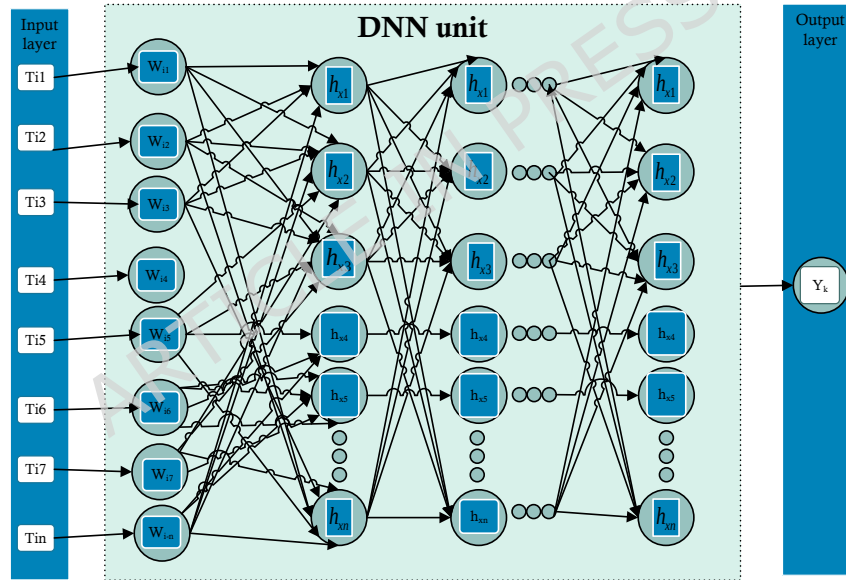


Figure 3: Designed model system based on the neurons.

4. LMB–DNN Workflow for the ABC Fractional Rabies Model

The proposed LMB–DNN framework for the eight-compartment, two-host rabies system is designed to efficiently approximate the fractional-order dynamics that describe the coupled human (P) and dog (D)

populations. The overall procedure consists of two major stages: fractional data generation and neural network training.

1. Supervised datasets are constructed by numerically solving the ABC fractional model for a set of parameter configurations and fractional orders $\alpha \in (0, 1)$. The fractional system is integrated on a normalized temporal domain $\tau \in [0, 1]$ using predictor–corrector methods based on the Grünwald–Letnikov and fractional ABM schemes. These solvers generate the reference state trajectories $\mathbf{y}(\tau) = (S_P, E_P, I_P, R_P, S_D, E_D, I_D, R_D)(\tau)$, which are then sampled to form input–target pairs $\{(\tau_n, \mathbf{y}(\tau_n))\}_{n=1}^N$. For consistency, the case $\alpha = 1$ can be verified using a classical ODE solver such as the Runge–Kutta method. All data are normalized to the range $[0, 1]$ (or $[-1, 1]$) prior to network training to improve numerical conditioning.
2. A feed-forward neural network is implemented in MATLAB using the `fitnet` function with the LM optimizer (`trainlm`). The network is designed with one hidden layer containing N_h neurons (typically $N_h = 10$), nonlinear activation functions in the hidden layer, and a linear activation in the output layer. The dataset is randomly partitioned into training (80%), validation (10%), and testing (10%) subsets. Training minimizes the mean squared error, and an early-stopping criterion based on validation performance is applied to prevent overfitting. This configuration achieves a good balance between accuracy and computational cost.

Figure 2 summarizes the overall LMB–DNN workflow adopted for the ABC fractional rabies model. The procedure starts with the numerical simulation of the fractional system, after which the resulting data are assembled, normalized, and partitioned into training, validation, and test sets. The neural network is then trained using the LMB algorithm, and its performance is evaluated through standard diagnostic tools, including MSE learning curves, regression plots, error histograms, and semilogarithmic AE profiles. Figure 3 provides a schematic illustration of the computation carried out by an individual neuron, highlighting the nonlinear activation in the hidden layer and the linear mapping at the output. All experiments are implemented programmatically using the MATLAB Deep Neural Network Toolbox through the `fitnet` interface. This scripted approach, rather than a graphical setup such as `nftool`, ensures full reproducibility and allows systematic sweeps over model parameters and fractional orders. The case-specific model parameters, including Λ_D and fractional order α for each simulation scenario, are listed in Table 1, while the training and validation metrics are summarized in Table 2.

5. Numerical Measures and Analysis

To assess the accuracy and robustness of the proposed LMB–DNN framework, we evaluate its performance against high-fidelity reference trajectories of the ABC fractional rabies system across three epidemiologically distinct parameter cases, summarized in Table 1 [31]. The inputs to the network are normalized time samples $\tau \in [0, 1]$, while the *targets* are the eight state components $\mathbf{y}(\tau) = (S_P, E_P, I_P, R_P, S_D, E_D, I_D, R_D)^\top$, computed using a fractional predictor–corrector scheme. Unless otherwise specified, the LMB–DNN employs two hidden layers with nonlinear activation functions and a linear output layer. The LM optimizer (`trainlm`) is used with an 80%/10%/10% division of data into training, validation, and testing subsets. All input and output signals are normalized to $[0, 1]$, and an early stopping mechanism halts training once validation loss ceases to improve.

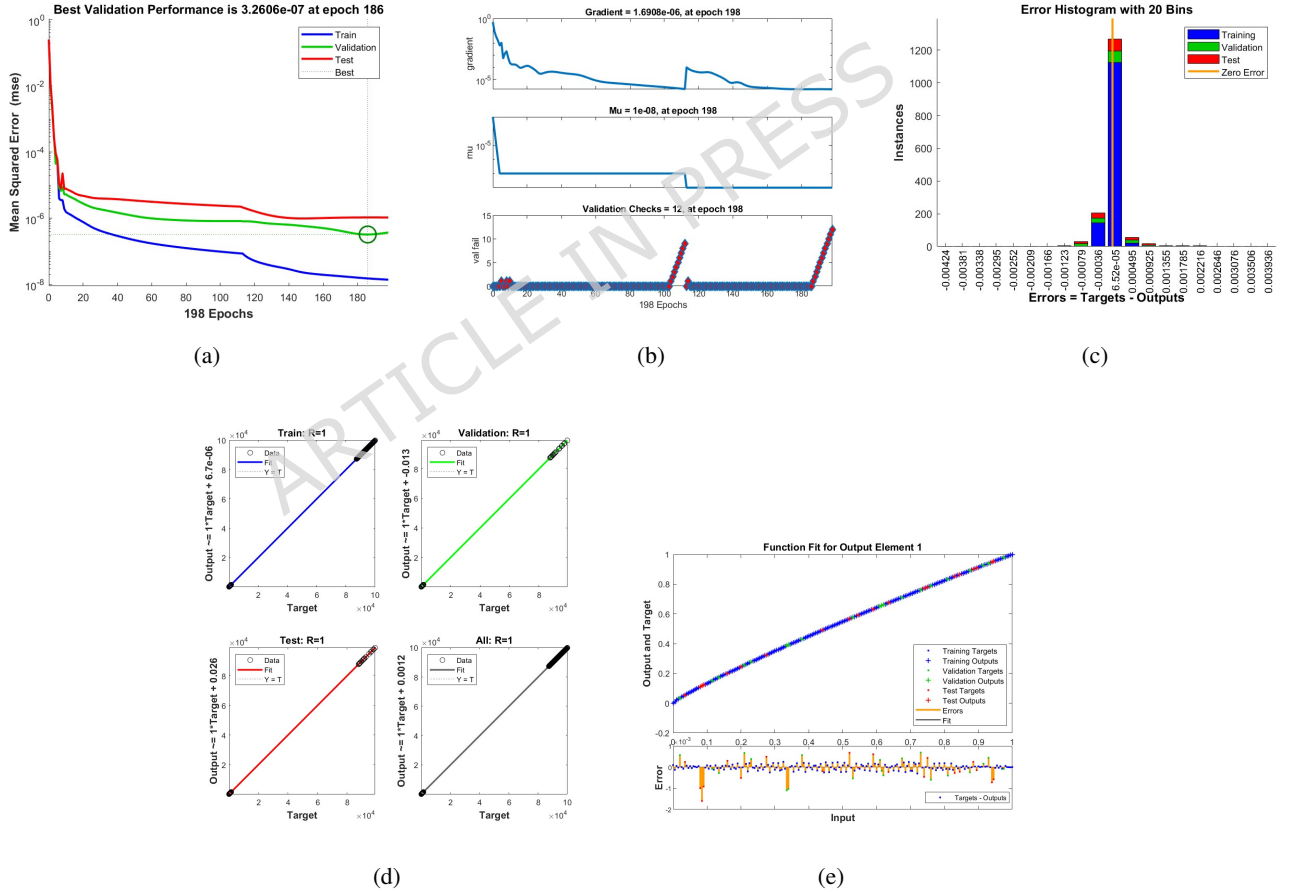


Figure 4: The model under consideration's statistical dynamics include (a) regression, (b) mean square error, (c) State transition, (d) error histogram, and (e) training fit for the LMB designed DNN for case 2.

Table 1: Parameters of the ABC fractional rabies model across the three test cases.

Case	Λ_P	μ_P	μ_D	α_P	α_D	m_P	m_D	B_{DD}	ε_P	ε_D	w_P	w_D	B_{DP}	Λ_D	α
Case 1	0.314	0.0074	0.056	1	1	0.03	0.01	1.58×10^{-7}	0.6	0.395	0.25	0.54	2.29×10^{-12}	4000	0.95
Case 2	0.314	0.0074	0.056	1	1	0.03	0.01	1.58×10^{-7}	0.6	0.395	0.25	0.54	2.29×10^{-12}	5600	0.90
Case 3	0.314	0.0074	0.056	1	1	0.03	0.01	1.58×10^{-7}	0.6	0.395	0.25	0.54	2.29×10^{-12}	7000	0.85

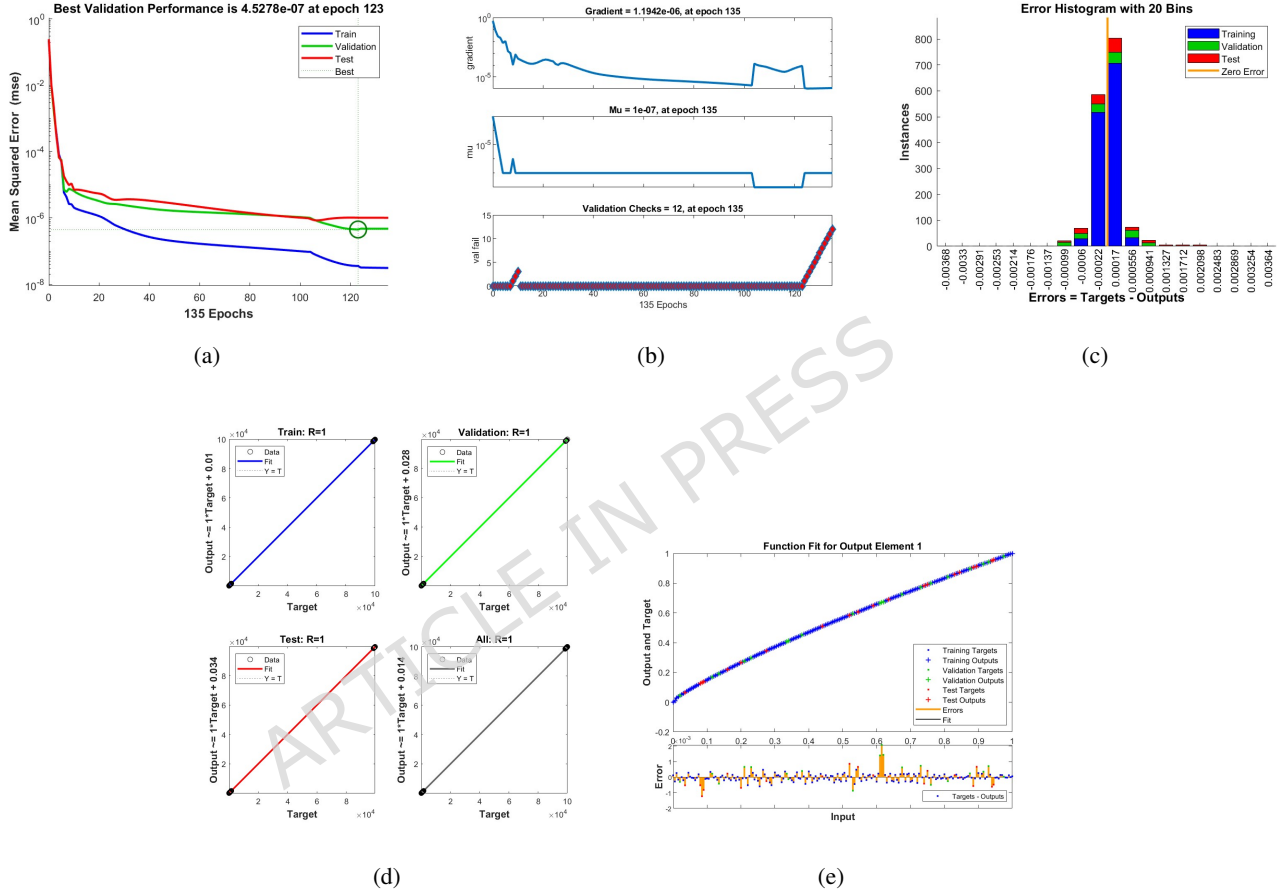


Figure 5: The model under consideration's statistical dynamics include (a) regression, (b) mean square error, (c) State transition, (d) error histogram, and (e) training fit for the LMB designed DNN for case 3.

The numerical simulations are designed to illustrate both the epidemiological behavior of the proposed ABC fractional rabies model and the performance of the LMB–DNN surrogate in approximating its dynamics. All figures are organized to separately highlight (i) model trajectories, (ii) learning accuracy, and (iii) error characteristics. For clarity, each figure focuses on a specific diagnostic aspect of the framework.

The reference solutions are obtained using a high-accuracy fractional ABM predictor-corrector scheme, while the corresponding neural outputs are produced by the proposed LMB-DNN model. Comparisons are performed across several epidemiological scenarios and different fractional orders in order to evaluate robustness and generalization.

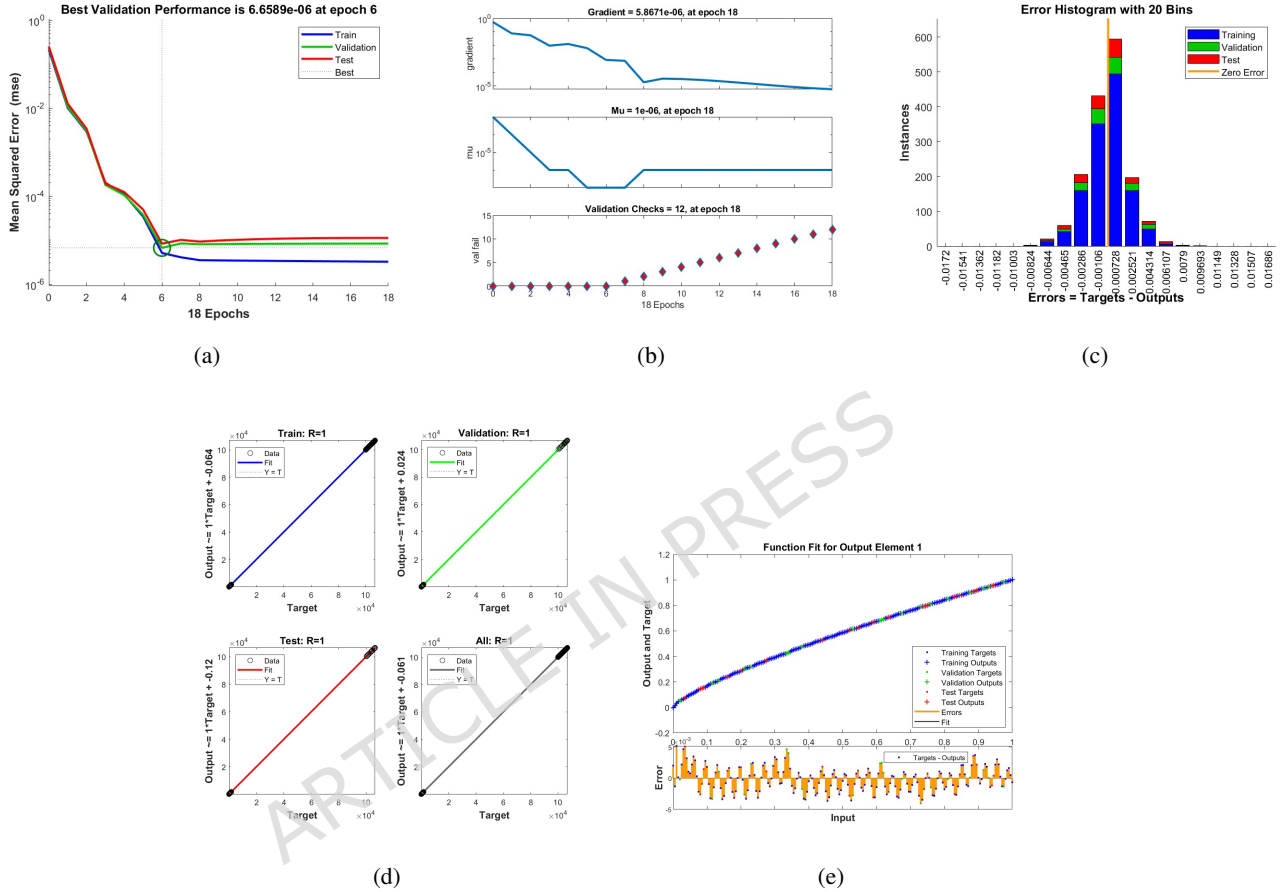


Figure 6: The model under consideration's statistical dynamics include (a) regression, (b) mean square error, (c) State transition, (d) error histogram, and (e) training fit for the LMB designed DNN for case 4.

Figures 4–6 summarize the main statistical diagnostics of the LMB-DNN for the considered cases. The regression plots show an almost one to one agreement between the neural predictions and the fractional reference solutions, indicating that the network successfully learns the underlying solution structure. The mean-squared-error (MSE) curves decrease smoothly during training and remain stable during the validation and testing stages, suggesting that overfitting is effectively avoided. In addition, the error histograms are tightly centered around zero, with small dispersion, which reflects unbiased residuals and stable numerical

learning. Taken together, these results demonstrate the accuracy, robustness, and reliability of the proposed LMB–DNN framework.

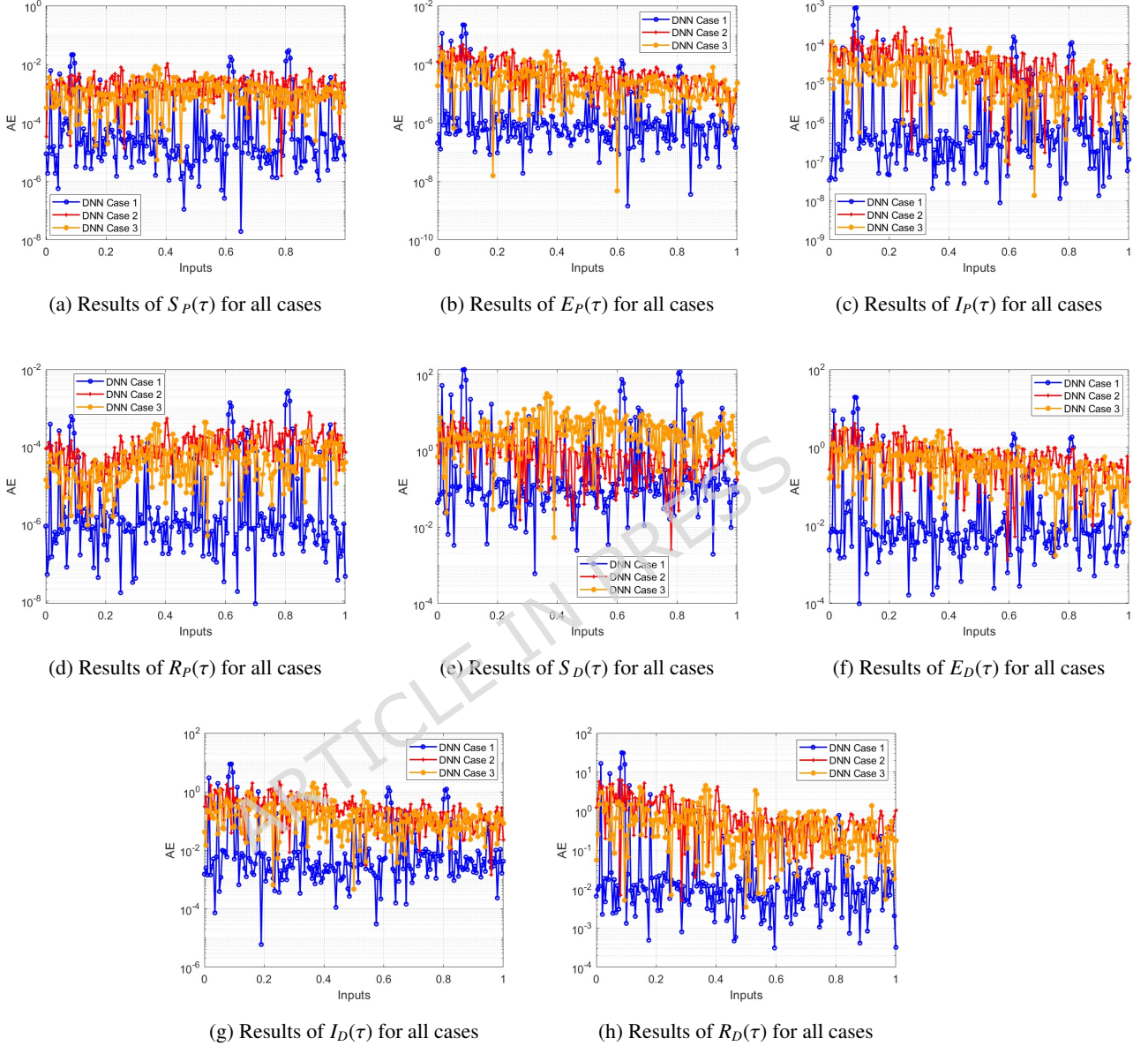


Figure 7: Comparison of the obtained results through LMB-DNN for each case of the model classes.

Figure 7 compares the LMB–DNN predictions with the trajectories generated by the Adam-based fractional solver. For every state component (S_P , E_P , I_P , R_P , S_D , E_D , I_D , R_D), the predicted and reference curves nearly coincide, reproducing both transient behavior and steady-state levels without noticeable phase

shifts or amplitude distortions. This close agreement highlights the accuracy and reliability of the trained surrogate.

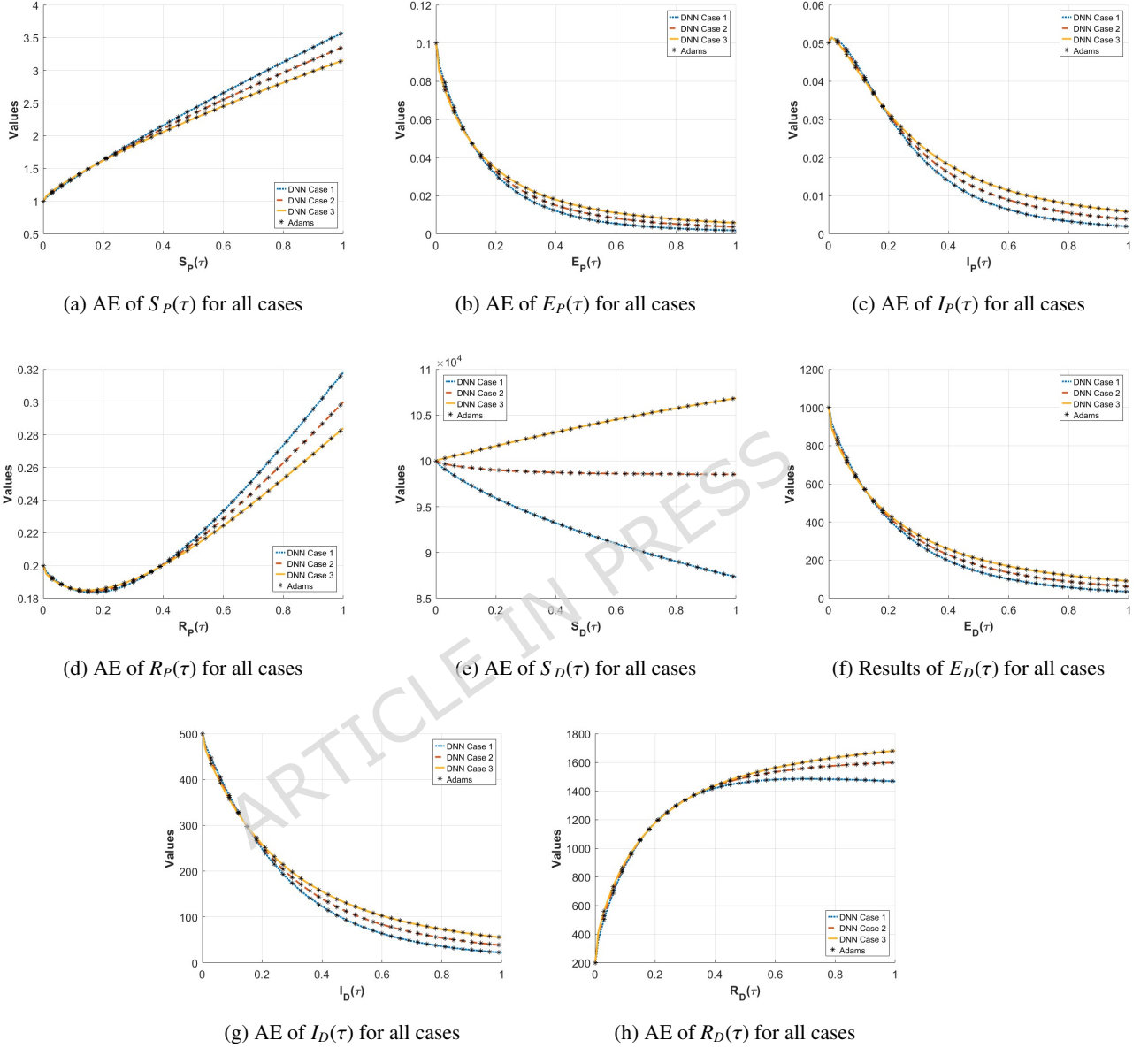


Figure 8: Absolute error (AE) based on the obtained results and Adam results via LMB-DNN.

Finally, the AE envelopes in Figure 8 display the pointwise differences between the LMB-DNN predictions and the reference solutions. The maximum AE across all state variables remains below 10^{-5} , with smooth, non-oscillatory profiles that indicate numerical stability and no overfitting; these small, structured

residuals confirm that the network captures the fractional-order memory and coupling effects in the ABC model.

Table 2: Training and performance statistics of the LMB–DNN for the ABC fractional rabies model (Cases 1–3).

Case	MSE			Performance	Gradient	μ	Epoch	Time (s)
	Train	Val	Test					
Case 1	1.5319×10^{-8}	3.2606×10^{-7}	1.0681×10^{-6}	3.2606×10^{-7}	1.7156×10^{-6}	1×10^{-8}	186	13.084
Case 2	3.5951×10^{-8}	4.5278×10^{-7}	1.0243×10^{-6}	4.5278×10^{-7}	8.6437×10^{-5}	1×10^{-8}	123	10.426
Case 3	5.0967×10^{-6}	6.6589×10^{-6}	8.3206×10^{-6}	6.6589×10^{-6}	8.6866×10^{-4}	1×10^{-7}	6	1.388

Table 2 reports the scalar metrics for training, validation, and testing across the three cases. The trained models achieve mean-squared-error values in the range 10^{-8} – 10^{-6} (Case 1: 1.5319×10^{-8} train, 3.2606×10^{-7} val, 1.0681×10^{-6} test; Case 2: 3.5951×10^{-8} , 4.5278×10^{-7} , 1.0243×10^{-6} ; Case 3: 5.0967×10^{-6} , 6.6589×10^{-6} , 8.3206×10^{-6}). Gradients are small (about 1.7×10^{-6} , 8.6×10^{-5} , and 8.7×10^{-4} for Cases 1–3), and the LM damping factors are stable ($\mu = 10^{-8}$ for Cases 1–2 and $\mu = 10^{-7}$ for Case 3). Validation and testing MSEs are closely matched, indicating strong generalization. Optimal validation performance was reached at roughly 186, 123, and 6 epochs, with wall-clock training times of about 13.08 s, 10.43 s, and 1.39 s for Cases 1–3, respectively, highlighting the efficiency and rapid convergence of the LM algorithm.

5.1. Effect of the fractional order α on rabies dynamics

The model is formulated using a fractional-order operator, it is essential to examine how the fractional order $\alpha \in (0, 1]$ influences the temporal evolution of rabies transmission. In fractional epidemic systems, α governs the strength of memory: smaller values of α correspond to stronger hereditary effects and longer influence of past states, whereas $\alpha \rightarrow 1$ recovers the classical (memoryless) integer-order dynamics. To assess the impact of memory, we simulate system (1.1) for multiple values of the fractional order in the interval $(0, 1]$, namely $\alpha = 0.70, 0.80, 0.90, 0.95$, and 1.00 , while keeping all other parameters fixed and using the same initial conditions. Figure 9 illustrates the trajectories of all compartments ($S_P, E_P, I_P, R_P, S_D, E_D, I_D, R_D$) for varying α .

The results demonstrate that changing α produces clear quantitative changes in the outbreak profiles. In particular, decreasing α modifies both transient and long-term behavior of the disease dynamics, typically leading to smoother trajectories, delayed outbreak timing, and a change in peak magnitude of the infectious

classes. These results indicate that the fractional order is more than a numerical tuning parameter; it has a clear biological interpretation related to memory effects in rabies transmission, including prolonged incubation, delayed behavioral responses, and persistence in infectious dynamics. From a control perspective, the dependence of the system dynamics on α suggests that both the timing and intensity of interventions may need to be adjusted when strong memory effects are present. As a result, incorporating fractional memory allows the model to represent rabies progression more realistically than classical integer-order formulations. A biological interpretation of the numerical effects associated with variations in the fractional order α is provided in Section 8.

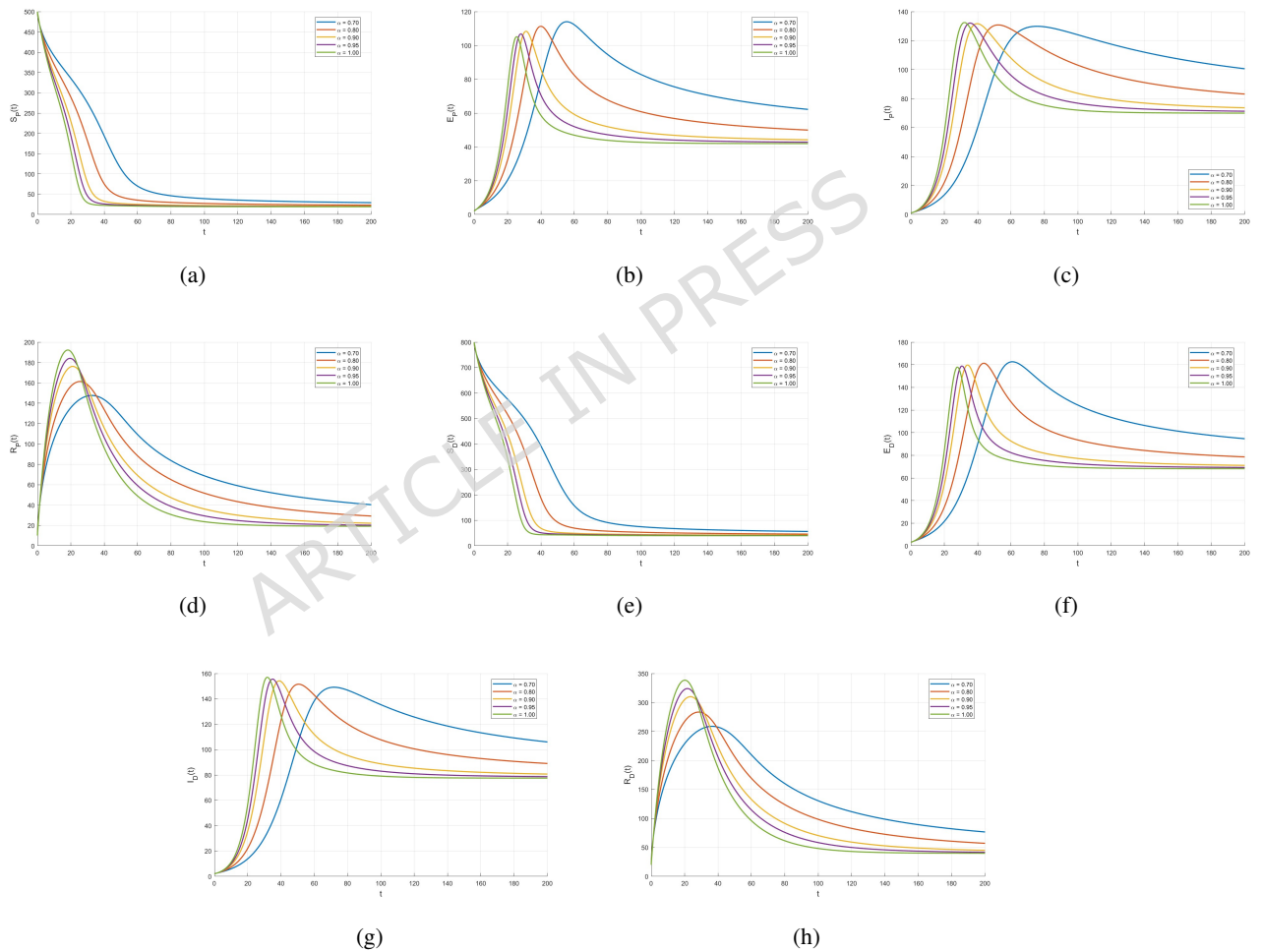


Figure 9: Time evolution of the human and dog compartments for different values of the fractional order $\alpha \in (0, 1]$. Lower values of α , corresponding to stronger memory effects, modify the peak intensity, delay the outbreak timing, and increase the persistence of infection relative to the classical case $\alpha = 1$.

6. Sensitivity Analysis

Sensitivity analysis is used to examine how changes in model parameters affect the behavior of a system. By varying one parameter at a time within a specified range while keeping the others fixed, it becomes possible to identify which parameters exert the strongest influence on the model output. This approach is particularly useful for understanding model behavior, assessing uncertainty, and highlighting parameters that are most relevant for control and intervention. For the proposed rabies transmission model, sensitivity analysis of the basic reproduction number R_0 indicates that the dog-to-dog transmission rate B_{DD} and the dog incubation parameter ϵ_D have a positive effect on R_0 , whereas the dog mortality rate μ_D and the removal rate α_D contribute negatively, as illustrated in Figure 10. From a public health perspective, these results suggest that control strategies focusing on dogs such as mass vaccination, rapid detection and removal of infectious animals, and reducing dog-to-dog contact are more effective than interventions targeting humans alone. This finding is consistent with existing rabies control guidelines and supports the epidemiological relevance of the proposed model.

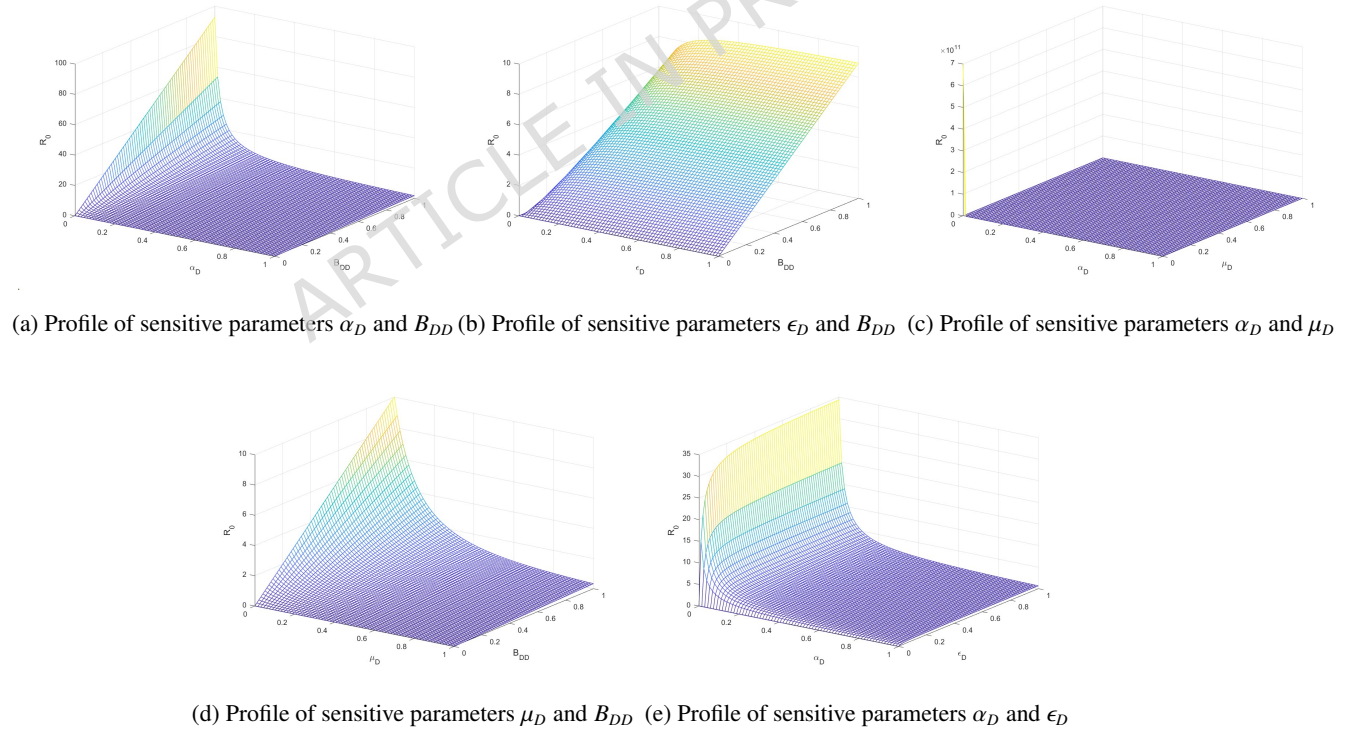


Figure 10: Sensitivity analysis of different parameters versus R_0 .

7. Computational Complexity and Training Efficiency Analysis

The computational behavior of the LM optimizer can be better understood by comparing it with the widely used Adam algorithm. The LM method incorporates curvature information through a Gauss–Newton–type update, which requires forming and solving a linear system at each iteration. As a result, its per-iteration complexity grows approximately as $O(n^3)$ for a network with n adjustable parameters. In contrast, Adam is a first-order gradient method whose update rules only involve simple vector operations, giving it a much lower per-iteration cost of $O(n)$. However, practical efficiency cannot be judged by per-step complexity alone. Because LM uses second-order information, it typically converges in far fewer iterations than Adam for smooth, well-behaved problems. This is exactly the setting encountered in the fractional rabies model, where the system dynamics are continuous and the neural approximation is compact.

The training times reported in Table 2 reflect this advantage. Despite the higher cost per update, LM reached its best validation performance within 186 epochs in Case 1 (13.08 s), 123 epochs in Case 2 (10.43 s), and only 6 epochs in Case 3 (1.39 s). The fast convergence suggests that the LM algorithm is able to locate a good solution without the long iteration counts usually required by first-order optimizers. In practical terms, this reduces the computational load, especially when the network must be retrained many times for parameter studies, uncertainty analysis, or scenario exploration. The method also delivers stable and accurate fits across all tested cases, as reflected in the regression results, the error histograms, and the consistently low MSE values. Taken together, these results indicate that LM achieves a good compromise between computational effort and convergence speed, making it a suitable choice for training neural models that describe fractional-order epidemic dynamics.

8. Biological Interpretation of the Fractional Order α

This section complements the numerical results presented in Section 5.1 by offering a biological interpretation of the fractional order α . In the context of rabies transmission, α should not be regarded simply as a mathematical tuning parameter, but rather as a quantity that reflects memory effects inherent to the disease process. When $\alpha \approx 1$, the system behaves similarly to a classical ordinary differential equation model, with infection progression and recovery occurring without appreciable memory. As α decreases below unity, the influence of past states becomes more pronounced, leading to slower dynamics, reduced infection peaks, and a spreading of transmission events over longer time intervals. From a practical standpoint, smaller values of α correspond to delays between exposure and the onset of active infection. This behavior is consistent

with the variable and often prolonged incubation period observed in rabies. The numerical results shown in Fig. 9, where the validation error is minimized for α in the range 0.90–0.95, further support the conclusion that moderate memory effects provide a more realistic representation of rabies dynamics.

Biologically, the fractional order can be interpreted as an aggregated description of several real-world mechanisms. These include persistent contact patterns within dog populations, the environmental survival of the virus on surfaces or in saliva, and immune memory processes that affect host susceptibility over time. Each of these introduces a temporal dependence that is difficult to capture using traditional integer-order models. By adjusting α , the proposed framework accounts for how past exposures influence current infection levels, resulting in simulated dynamics that better align with observed rabies epidemiology. Overall, the fractional modeling approach offers both numerical flexibility and meaningful biological insight by embedding memory-driven effects directly into the transmission structure.

9. Limitations and Future Work

This study offers a detailed analysis of a fractional-order rabies transmission model and demonstrates the usefulness of the LMB–DNN framework, several limitations should be noted. First, the proposed model does not include spatial structure. As a result, it does not explicitly account for dog movement, territorial behavior, or transmission between neighboring communities, all of which can influence the spread of rabies. Incorporating spatial diffusion, metapopulation frameworks, or simple movement rules would improve the biological realism of the model and allow spatial heterogeneity in disease dynamics to be explored. Second, the model is formulated as a deterministic system of fractional differential equations. In real epidemiological settings, randomness arising from demographic variability, heterogeneous contact patterns, and environmental fluctuations can substantially affect outbreak trajectories, particularly during the early stages of an epidemic or in regions with low incidence. Extending the present framework to stochastic or hybrid deterministic–stochastic formulations would provide a more realistic description of rabies transmission in the presence of uncertainty.

The current formulation also assumes a constant dog recruitment rate, whereas free-roaming dog populations often exhibit seasonal birth pulses. Incorporating seasonal forcing or time-dependent demographic processes could help capture the cyclical or irregular outbreak patterns commonly observed in rabies surveillance data. In addition, control measures are modeled through fixed parameters. Future studies could consider optimal or adaptive intervention strategies, such as time-varying vaccination coverage or feedback–

based control policies, to increase the relevance of the model for public–health planning. From a computational perspective, the neural network surrogate is trained on temporal trajectories generated for selected parameter configurations. As a result, its predictive accuracy outside the sampled parameter space cannot be guaranteed. Future work could focus on extending the training domain or developing surrogate models that more directly learn the full parameter space. In this regard, physics–informed neural networks (PINNs) offer a promising alternative, as they can be trained directly on the governing fractional equations without relying exclusively on pre–computed datasets. Furthermore, Bayesian neural networks or ensemble learning techniques could be incorporated to provide explicit uncertainty quantification, which is particularly important for epidemiological decision–making.

Another limitation of the present study is the lack of direct validation using reported rabies incidence data. In practice, epidemiological surveillance data are often subject to underreporting, delays in reporting, and measurement noise, which makes model calibration and evaluation challenging. For this reason, the proposed neural surrogate is first validated against a reliable fractional numerical solver, allowing the numerical accuracy and learning performance of the framework to be assessed independently of data–related uncertainties.

Future work will focus on incorporating real epidemiological data, such as time series of human rabies cases, records of dog bites, vaccination coverage, and estimates of dog population size. These data can be integrated through parameter estimation and data–assimilation approaches. For instance, the neural surrogate could be combined with calibration procedures based on least–squares fitting, Bayesian inference, or ensemble filtering to align model outputs with observations while preserving the underlying mechanistic structure of the fractional model. Such extensions would improve the realism and practical applicability of the proposed framework, and enhance its relevance for public–health decision–making, while retaining the benefits of fractional memory modeling and computational efficiency demonstrated in this study.

10. Conclusions

In this study, we investigated an eight–compartment rabies transmission model describing interactions between human and dog populations, formulated using the ABC fractional derivative. To efficiently approximate the resulting memory–dependent dynamics, we introduced an integrated intelligent–computing framework that couples a LMB–DNN. High–accuracy reference solutions were obtained using a fractional ABM predictor–corrector scheme on a normalized time interval, and the neural surrogate was trained using

an 80%/10%/10% split for training, validation, and testing. A compact baseline network architecture was adopted, while deeper configurations were also explored to examine robustness. The numerical results show close agreement between the LMB–DNN predictions and the corresponding fractional reference solutions across all eight state variables. This agreement is supported by several diagnostic measures, including regression plots with near-unity correlation, smoothly decreasing MSE curves with effective early stopping, error histograms centered near zero with small variance, and semilogarithmic absolute-error profiles that remain low throughout the simulation period. Together, these indicators demonstrate that the LMB-trained network is able to reproduce the ABC fractional rabies dynamics accurately while maintaining numerical stability and good generalization. Beyond numerical accuracy, the proposed approach contributes to ANN-based epidemic modeling by combining biological realism with computational efficiency. The fractional order naturally captures memory-driven features of rabies transmission, such as prolonged incubation and delayed responses, while the harmonic-mean incidence formulation accounts for saturation effects in host contacts. At the same time, the LMB–DNN surrogate provides a computationally efficient alternative to repeated fractional simulations, enabling rapid exploration of scenarios, sensitivity analyses, and decision-oriented investigations. A hyperparameter sensitivity study further confirms the robustness of the framework. Validation errors remain low over wide ranges of the LM damping parameter and network width, with optimal performance typically observed around $\mu \approx 10^{-4}$ and fractional orders $\alpha \in [0.90, 0.95]$. These results suggest that relatively compact neural architectures are sufficient to capture the smooth, memory-driven dynamics of the ABC fractional model, avoiding unnecessary over-parameterization while preserving high predictive accuracy. The proposed LMB–DNN framework offers an accurate, efficient, and reliable surrogate modeling strategy for fractional epidemic systems with harmonic-mean incidence and multi-host coupling. While the present study focuses on rabies transmission, the methodology is readily extensible to other multi-host or multi-scale fractional epidemic models, as well as to broader scientific applications where memory effects and nonlocal dynamics play a central role, including fractional process modeling, bioinformatics with temporal dependence, and complex transport–reaction systems. Its MATLAB based implementation supports transparent parameter sweeps, reproducible training, and rapid what if analyses, making it a practical and flexible tool for researchers studying fractional-order infectious-disease dynamics and control.

Availability of data and material

The datasets used and/or analyzed during the current study available from the corresponding author on reason-able request.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author's Contributions

R.S. and I. conceptualized the study; R. S. designed the methodology; A. A. and A. M. D. was responsible for software; R. S. was responsible for formal analysis; R. S. and I. were responsible for the investigation of the study; R. S and I. was responsible for the resources; R. S. wrote the original draft; R. S. was responsible for reviewing and editing; R. S. supervised the study; A. A. was responsible for project administration; A. M. D. was responsible for funding acquisition. All authors have read and approved the final version of the manuscript for publication.

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